

Sankalp Pandey

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Education

University of Arkansas

Aug. 2022 – May 2026

B.S. in Computer Science & B.S. in Computer Engineering

Minors: Mathematics, Data Analytics

Honors College Ambassador

GPA: 3.77/4.0

Research Experience

Research Assistant

Fayetteville, AR

University of Arkansas, Computer Vision and Image Understanding Lab

Feb. 2024 – Present

Advisor: Dr. Khoa Luu

- Conducting research in multimodal reasoning for brain signal decoding and quantum material discovery to develop scalable AI systems for scientific discovery.

Publications

TIMBRE: Time-aware Multimodal Vision-Brain Encoding

Xuan-Bac Nguyen, Sankalp Pandey, Suhyun Kim, Hojin Jang, Arabinda Kumar Choudhary, Khoa Luu

Under review

φ -Adapt: A Physics-Informed Adaptation Learning Approach to 2D Quantum Material Discovery

Hoang-Quan Nguyen, Xuan-Bac Nguyen, Sankalp Pandey, Tim Faltermeier, Nicholas Borys,

Hugh Churchill, Khoa Luu

Under review

QuPAINT: Physics-Aware Instruction Tuning Approach to Quantum Material Discovery

Xuan-Bac Nguyen, Sankalp Pandey, Suhyun Kim, Hojin Jang, Arabinda Kumar Choudhary, Khoa Luu

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Findings, 2026

CLIFF: Continual Learning for Incremental Flake Features in 2D Material Identification

Sankalp Pandey, Xuan-Bac Nguyen, Nicholas Borys, Hugh Churchill, Khoa Luu

Neural Information Processing Systems (NeurIPS) AI for Accelerated Materials Discovery Workshop, 2025

QMoE: A Quantum Mixture of Experts Framework for Scalable Quantum Neural Networks

Hoang-Quan Nguyen, Xuan-Bac Nguyen, Sankalp Pandey, Samee U. Khan, Ilya Safro, Khoa Luu

IEEE International Conference on Quantum Computing and Engineering (QCE) QCRL Workshop, 2025

Industry Experience

Software Engineer Intern

Springdale, AR

Hidalga Technologies

Aug. 2024 – Aug. 2025

- Developed a gradient boosting model with 87.6% accuracy on 30K+ patient cases to predict prior authorization outcomes.
- Orchestrated a multi-agent system for producing synthetic FHIR patient records and simulating insurance coverage outcomes with automated approval/denial reasoning.
- Piloted an automated document parsing and validation system using Azure OpenAI, tested on 250+ forms and reducing turnaround time by 64% compared to manual processing.

Teaching Experience

Teaching Assistant

Jan. 2024 – May 2025

University of Arkansas, Department of EECS, Fayetteville, AR

- CSCE 20104 (Programming Foundations II)
- CSCE 20004 (Programming Foundations I)

Competitions

Algonauts Challenge 2025 – 5th / 67 teams

Jul. 2025

- International benchmark on predicting brain responses to naturalistic video stimuli.
- Engineered multimodal embeddings integrating visual, audio, and contextual features to improve alignment between neural signals and video content.

San Francisco Crime Classification – 153rd / 2,332 teams

Dec. 2024

- Kaggle challenge on classifying incidents into 39 crime categories using spatio-temporal features from historic SFPD crime reports.
- Trained a CatBoost model on 800K+ records, achieving a weighted F1 score of 0.2505.
- Ranking based on score compared to frozen leaderboard

Projects

QuantumFlake [↗](#)

May 2025 – Present

- Published an open-source library that automates flake detection and layer-count classification, accelerating quantum materials research workflows.
- Achieved a 0.94 weighted F1-score for the ResNet-18 classifier and processing images at a rate of 0.11s/image.
- Increased accessibility for the research community by publishing pre-trained models on HuggingFace, developing a live demo, and creating a one-line command-line interface for rapid analysis.

Grocify [↗](#)

Feb. 2025 – Mar. 2025

- Built a Streamlit app to automate grocery shopping by converting text recipes or meal images into prefilled Walmart carts, leveraging ViT, Spoonacular API, and OpenAI GPT for precise ingredient extraction.
- Delivered a fully automated pipeline with average end-to-end processing time ≤ 8 seconds.

FlowForge [↗](#)

Feb. 2025

- Built a mobile application using React Native and TypeScript to for mindfulness and productivity.
- Integrated Google Gemini API to for context-aware AI assistants and motion sensors to detect phone usage.

Honors & Awards

University of Arkansas Honors College Conference Travel Grant

Aug. 2025

University of Arkansas Honors College Research Grant

Aug. 2024

Arkansas Governor's Distinguished Scholarship

Aug. 2022

University of Arkansas Chancellor's Scholarship

Aug. 2022

Skills

Languages: Python, Java, C, C++, C#, JavaScript/TypeScript, SQL, PHP, Bash

Libraries and Frameworks: TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, React, Next.js, SvelteKit

Tools and Platforms: AWS, Microsoft Azure, Docker, GitHub Actions, MySQL, PostgreSQL, Vercel

Research Interests: Computer Vision, Multimodal Learning, Materials Discovery, Computational Neuroscience

References

Available upon request.